Smallest Polyomino Containing Every Free n-Omino: A327094 for n up to 10

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Pick a size n and write down every free n-omino – every connected shape of n squares, counted once per rotation and reflection. Now ask for the smallest single polyomino that hides every one of those shapes somewhere inside it. The number of cells in that smallest universal container is $a(n)$, and this note pins down $a(1)$ through $a(10)$.

The problem

A *free n-omino* is a connected set of n unit cells on the square grid, taken up to rotation and reflection. There are $1, 1, 2, 5, 12, 35, 108, 369, 1285, 4655$ of them for $n = 1$ through 10. Call a polyomino a *container* for size n if every free n-omino fits inside it under some rotation, reflection, and translation. We want $a(n)$, the fewest cells a container for size n can have. This is OEIS A327094 (Peter Kagey, 2019), who gave the upper bound $a(n) \leq n(n-1)/2$ for $n \geq 4$ from a right-triangular staircase.

The values listed here are proved minimal among containers whose bounding box lies within a searched window: height at least $\lceil n/2 \rceil$, width at least n, and area at most twice the cell count. A smaller container with a more elongated bounding box outside that window is not separately excluded, so global minimality stays conjectural. Within the window the values are exact.

Definitions

We work on the square lattice $\mathbb{Z}^2$ with edge (4-neighbour) adjacency. A *polyomino* is a finite, edge-connected set of cells. The dihedral group D_4 (four rotations and four reflections) acts on polyominoes; two polyominoes are the same *free* polyomino when one is a D_4 image of a translate of the other.

A polyomino C *contains* a free n-omino P when some D_4 image of P, translated, is a subset of C. C is a *container* for size n when it contains every free n-omino. Then $a(n)$ is the smallest cell count of such a container. The first two terms are immediate: $a(1) = 1$ (the single monomino) and $a(2) = 2$ (the single domino).

A candidate container of k cells is sought inside a finite set of bounding boxes, the *window*: boxes of height h and width w with $h \geq \lceil n/2 \rceil$ (a container must span the straight n-omino in its short direction), $w \geq n$ (it must span the straight n-omino lengthwise), and $k \leq h \cdot w \leq 2k$. The height and width bounds are forced; the area cap is the one assumption that makes the result conditional.

The values

Result. For $n = 1$ through 10,

n	1	2	3	4	5	6	7	8	9	10
a(n)	1	2	4	6	9	12	17	20	26	31

Each value is the smallest cell count over all containers whose bounding box lies within the window above: a container of $a(n)$ cells exists, and no container of $a(n) - 1$ cells exists within that window. Kagey's bound $n(n-1)/2$ is tight only at $n = 3$ and $n = 4$ and loosens after that (45 against 31 at $n = 10$).

How we know

Each term needs an upper bound and a matching lower bound. We search top-down from Kagey's ceiling $\max(n(n-1)/2, n+1)$. For a candidate k we ask whether a connected k-cell container exists; the last k that answers yes is $a(n)$.

A constraint solver supplies the upper bound: for each n it finds an explicit $a(n)$-cell container that holds all of the free n-ominoes at once. That witness is a concrete shape, so the upper bound needs no trust in the solver.

A SAT solver supplies the lower bound by refutation. At $k = a(n) - 1$ it proves that no k-cell subset of any window grid – connected or not – can hold every piece, using piece-driven counterexample-guided refinement to add only the clauses that matter. Dropping the connectivity requirement only makes the problem easier, so "no k-cell subset works" is strictly stronger than "no connected k-cell container works"; the lower bound follows without any connectivity argument.

The values are confirmed two independent ways. A from-scratch geometric check re-enumerates the free n-ominoes with its own code and confirms by direct set inclusion that the reported container really holds every one – an algorithmic check of the upper bound that shares nothing with the search. And the refutation is replayed as a machine-checked certificate and re-run on a second, unrelated SAT engine, so the lower bound does not rest on one solver being bug-free. Removing each search heuristic in turn (symmetry breaking, lonely-cell clauses, the translation breaker) leaves every value unchanged, so the heuristics only prune the search and never steer the answer.

Examples

Cells are marked `#`, empty cells `.`. For $n = 3$, $a(3) = 4$:

```
#..
###
```

For $n = 5$, $a(5) = 9$, a cross-like container:

```
###..
#####
..#..
```

For $n = 7$, $a(7) = 17$:

```
...#...
.####..
#######
#####..
```

And the largest proved term, $n = 10$, $a(10) = 31$, a staircase that thickens toward its base:

```
.......###
......####
....######
##########
..########
```

The shapes grow chunkier as n rises: the early containers are nearly a single fat row, while $a(10)$ needs several stacked rows to absorb the more contorted decominoes. The figure gallery shows the smallest container for every n.

Patterns and open questions

The sequence grows roughly quadratically but never smoothly. Its first differences are $1, 2, 2, 3, 3, 5, 3, 6, 5$ and its second differences are $1, 0, 1, 0, 2, -2, 3, -1$, dropping back at $n = 7 \to 8$. That irregularity is real, not noise: a least-squares quadratic (about $0.269n^2 + 0.399n + 0.25$) tracks $n \leq 6$ but misses $n = 7$ (predicting 16, not 17) and $n = 8$ (predicting 21, not 20); a cubic fails the same way; no small-integer recurrence $a(n) = p\, a(n-1) + q\, a(n-2) + s$ fits exactly; and a floor form $\lfloor (n^2 + c)/d \rfloor + \delta$ matches fewer than seven of the ten terms for every small c and d. We found no closed form over the proved range.

The frontier sits at $n = 10$. The obstacle is the piece count, which jumps from 4655 at $n = 10$ to 17073 at $n = 11$; the model for $n = 11$ exhausted memory within the per-term budget before settling a value. Kagey's conjectured $a(11) = 37$ and $a(12) = 43$ lie just past where our search reaches, so they stay open here. Reaching $a(11)$ likely needs a more compact encoding of the placement constraints, or a decomposition that avoids holding all 17073 pieces at once.

Two larger questions remain. First, whether any $a(n)$ admits a closed form: ten irregular terms argue against it but cannot rule out a formula that only surfaces later. Second, whether within-window minimality equals global minimality. The window's height and width bounds are forced; only the area cap is assumed. Settling whether a minimal container can ever be more elongated than that cap – by proof or by counterexample – would close the gap.

A related sequence, A352029 (John Mason, 2022), counts the minimalist containers rather than measuring them; it is a different quantity, noted here only for context.

Further reading

The paper and the figure gallery are at https://github.com/pexley-math/oeis-A327094.

This work was inspired by the OEIS and the community of contributors who maintain it.

- Kagey, P. (2019). "A327094: Smallest polyomino containing all free n-ominoes." The On-Line Encyclopedia of Integer Sequences. https://oeis.org/A327094
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- Redelmeier, D. H. (1981). "Counting Polyominoes: Yet Another Attack." Discrete Mathematics 36, 191-203.
- Sloane, N. J. A. "A000105: Number of free polyominoes (or square animals) with n cells." The On-Line Encyclopedia of Integer Sequences. https://oeis.org/A000105